

Path Independence

$$\int_C \vec{F} \cdot \hat{T} \, ds = \int_C a \, ds = a \int_C ds$$

or computing

$$\int_C -y \, dx + x \, dy$$

$$x = a \cos \theta, \quad y = a \sin \theta, \quad 0 \leq \theta \leq 2\pi$$

$$\int_0^{2\pi} a^2 (\sin^2 \theta + \cos^2 \theta) \, d\theta$$

$$\vec{F} = \langle M, N \rangle$$

$$\int_C \vec{F} \cdot d\vec{r}$$

We need to evaluate

$$\int_C y \, dx + x \, dy$$

1. Along the x-axis (C_1)

Path from $(0, 0)$ to $(1, 0)$.

$$y = 0, \quad dy = 0$$

Therefore,

$$\int_{C_1} y \, dx + x \, dy = \int_{C_1} 0 \, dx + 0 = 0$$

Geometrically, on the x-axis,

$$\vec{F} \parallel \hat{j}$$

hence

$$\vec{F} \cdot \hat{T} = 0$$

2. Portion of the Unit Circle (C_2)

Parameterize the curve by

$$x = \cos \theta, \quad y = \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{4}$$

Then

$$dx = -\sin \theta \, d\theta, \quad dy = \cos \theta \, d\theta$$

Substitute into the integral:

$$\begin{aligned} \int_{C_2} y \, dx + x \, dy &= \int_0^{\pi/4} [\sin \theta (-\sin \theta) + \cos \theta (\cos \theta)] \, d\theta \\ &= \int_0^{\pi/4} (\cos^2 \theta - \sin^2 \theta) \, d\theta \end{aligned}$$

Using the identity

$$\cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

we get

$$\begin{aligned} &= \int_0^{\pi/4} \cos 2\theta \, d\theta \\ &= \left[\frac{1}{2} \sin 2\theta \right]_0^{\pi/4} \\ &= \frac{1}{2} \end{aligned}$$

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Total Work

$$\begin{aligned} \oint_C &= \int_{C_1} + \int_{C_2} + \int_{C_3} \\ &= 0 + \frac{1}{2} - \frac{1}{2} = 0 \end{aligned}$$

Special Case

Suppose

$$\vec{F} = \nabla f$$

Then $f(x, y)$ is called a **potential function**.

We can simplify the evaluation of

$$\int_C \vec{F} \cdot d\vec{r}$$

using the **Fundamental Theorem of Calculus for Line Integrals**.

Fundamental Theorem for Line Integrals

$$\int_C \nabla f \cdot d\vec{r} = f(P_1) - f(P_0)$$

Equivalently,

$$\int_C f_x dx + f_y dy = \int_C df = f(P_1) - f(P_0)$$

Proof

$$\int_C \nabla f \cdot d\vec{r} = \int_C f_x dx + f_y dy$$

Let

$$x = x(t), \quad y = y(t)$$

Then

$$dx = x'(t) dt, \quad dy = y'(t) dt$$

Therefore,

$$\begin{aligned} \int_C \nabla f \cdot d\vec{r} &= \int_C \left(f_x \frac{dx}{dt} + f_y \frac{dy}{dt} \right) dt \\ &= \int_{t_0}^{t_1} \frac{df}{dt} dt \end{aligned}$$

$$\int_{t_0}^{t_1} \frac{df}{dt} dt = [f(x(t), y(t))]_{t_0}^{t_1} = f(P_1) - f(P_0)$$

Example

Let

$$\vec{F} = \langle y, x \rangle$$

Then

$$\nabla f = \vec{F}, \quad f(x, y) = xy$$

So,

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) - f(1, 0) \\ &= \frac{1}{2} - 0 = \frac{1}{2} \end{aligned}$$

Warning

Everything today only applies if $\vec{F}$ is a gradient field.

Otherwise, these results do not necessarily hold.

Consequences of the Fundamental Theorem

If $\vec{F}$ is a gradient field, then line integrals are path-independent.

Path Independence

If C_1 and C_2 have the same start and end points, then

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$$

Conservative Vector Fields

If

$$\vec{F} = \nabla f$$

then $\vec{F}$ is called **conservative**.

For a closed curve C ,

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

Indeed,

$$\oint_C \nabla f \cdot d\vec{r} = f(\text{end}) - f(\text{start}) = 0$$

since the curve is closed.

Remark

Let

$$\vec{F} = \langle -y, x \rangle$$

and let C be the unit circle.

Then

$$\int_C \vec{F} \cdot \hat{T} \, ds \neq 0$$

because

$$\vec{F} \parallel \hat{T}$$

and

$$\vec{F} \cdot \hat{T} = |\vec{F}| = 1$$

Hence, $\vec{F}$ is:

- not conservative,
- not path independent,
- not a gradient field.

Physics Interpretation

If a force field $\vec{F}$ is a gradient of a potential function,

$$\vec{F} = \nabla f$$

then the work done by $\vec{F}$ equals the change in the value of the potential.

Examples include:

- gravitational fields,
- electrical fields.

Conservativeness

Conservativeness means no energy can be extracted from the field for free.

Total energy is conserved.

Equivalent Properties

The following statements are equivalent:

1. $\vec{F}$ is conservative.

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

for all closed curves C .

- 2.

$$\int_C \vec{F} \cdot d\vec{r}$$

is path independent.

3. $\vec{F}$ is a gradient field.

$$\vec{F} = \nabla f$$

for some potential function $f(x, y)$.

- 4.

$$M dx + N dy$$

is an exact differential:

$$M dx + N dy = df$$

Gradient Fields and Potential Functions

If

$$\vec{F} = \nabla f$$

is a gradient field, then

$$\int_C \vec{F} \cdot d\vec{r} = f(P_1) - f(P_0)$$

Path Independence and Conservative Fields

A vector field is conservative if

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

for every closed curve C .

This is equivalent to path independence.

Testing Whether $\vec{F} = \langle M, N \rangle$ is a Gradient Field

Suppose

$$\vec{F} = \nabla f$$

Then

$$M = f_x, \quad N = f_y$$

Since mixed partial derivatives are equal,

$$f_{xy} = f_{yx}$$

we obtain

$$M_y = N_x$$

Conversely, if

$$M_y = N_x$$

and $\vec{F} = \langle M, N \rangle$ is defined and differentiable everywhere, then $\vec{F}$ is a gradient field.

Example

Let

$$\vec{F} = \langle -y, x \rangle$$

Then

$$M = -y, \quad N = x$$

Compute the partial derivatives:

$$\frac{\partial M}{\partial y} = -1$$

and

$$\frac{\partial N}{\partial x} = 1$$

Since

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

the vector field $\vec{F}$ is not a gradient field.

Example

Consider the vector field

$$\vec{F} = (4x^2 + axy)\hat{i} + (3y^2 + 4x^2)\hat{j}$$

Let

$$M = 4x^2 + axy, \quad N = 3y^2 + 4x^2$$

We determine the value of a for which $\vec{F}$ is a gradient field.
Compute:

$$M_y = ax$$

and

$$N_x = 8x$$

For a gradient field, we require

$$M_y = N_x$$

Hence,

$$ax = 8x$$

so

$$a = 8$$

Finding the Potential Function

Once $M_y = N_x$, we can find a potential function $f(x, y)$.

Computing Line Integrals

If

$$\vec{F} = \nabla f$$

then

$$\int_C \vec{F} \cdot d\vec{r} = f(x, y) - f(0, 0)$$

Equivalently,

$$f(x, y) = \int_C \vec{F} \cdot d\vec{r} + f(0, 0)$$

Substituting $a = 8$,

$$\vec{F} = \langle 4x^2 + 8xy, 3y^2 + 4x^2 \rangle$$

On C_1 ,

$$x : t \text{ from } 0 \text{ to } x, \quad y = 0, \quad dy = 0$$

Computing the Potential Function

On C_1 ,

$$\begin{aligned} \int_{C_1} \vec{F} \cdot d\vec{r} &= \int_0^x 4x^2 dx \\ &= \left[\frac{4x^3}{3} \right]_0^x = \frac{4x^3}{3} \end{aligned}$$

On C_2 , let y vary from 0 to y .

$$\begin{aligned} \int_{C_2} \vec{F} \cdot d\vec{r} &= \int_0^y (3y^2 + 4x^2) dy \\ &= \left[\frac{3y^3}{3} + 4x^2y \right]_0^y \\ &= y^3 + 4x^2y \end{aligned}$$

Therefore,

$$f(x, y) = \frac{4x^3}{3} + y^3 + 4x^2y + C$$

Method 2: Antiderivatives

We want to solve

$$f_x = 4x^2 + 8xy \quad (1)$$

and

$$f_y = 3y^2 + 4x^2 \quad (2)$$

Finding the Potential Function

Suppose

$$f_x = 4x^2 + 8xy$$

Integrating with respect to x ,

$$f = \frac{4x^3}{3} + 4x^2y + g(y)$$

Differentiate with respect to y :

$$f_y = 4x^2 + g'(y)$$

Match this with

$$f_y = 3y^2 + 4x^2$$

Thus,

$$4x^2 + g'(y) = 3y^2 + 4x^2$$

so

$$g'(y) = 3y^2$$

Integrating,

$$g(y) = y^3 + C$$

Therefore,

$$f = \frac{4x^3}{3} + 4x^2y + y^3 + C$$

Conservative Vector Fields

A vector field

$$\vec{F} = \langle M, N \rangle$$

is a gradient field in a region of the plane if it is conservative.

For a closed curve C ,

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

Also,

$$N_x = M_y$$

at every point.

If $\vec{F}$ is defined on the entire plane, or on a simply connected region, then this condition implies that $\vec{F}$ is conservative.

Curl and Conservative Fields

Definition: For a vector field

$$\vec{F} = \langle M, N \rangle,$$

the curl is defined by

$$\text{curl}(\vec{F}) = N_x - M_y.$$

To test for conservativeness:

$$\text{curl}(\vec{F}) = 0.$$

For a velocity field, curl measures the rotational component of the flow.

If

$$\vec{F} = \langle a, b \rangle,$$

then

$$\text{curl}(\vec{F}) = \frac{\partial b}{\partial x} - \frac{\partial a}{\partial y}.$$

Example

Let

$$\vec{F} = \langle -y, x \rangle.$$

Then

$$\text{curl}(\vec{F}) = \frac{\partial(x)}{\partial x} - \frac{\partial(-y)}{\partial y} = 1 - (-1) = 2.$$

Thus,

$$\operatorname{curl}(\vec{F}) = 2.$$

Curl measures angular velocity or the rotational component of a velocity field.

For a force field, curl measures the torque exerted on a test object in the field.

$$\frac{\text{torque}}{\text{moment of inertia}} = \frac{d}{dt}(\text{angular velocity})$$

Similarly,

$$\frac{\text{force}}{\text{mass}} = \frac{d}{dt}(\text{velocity}).$$

Lecture 22: Green's Theorem

Recall:

$$\operatorname{curl}(\vec{F}) = N_x - M_y$$

where

$$\vec{F} = \langle M, N \rangle.$$

We want to compute

$$\oint_C \vec{F} \cdot d\vec{r}.$$

Green's Theorem

If C is a closed curve enclosing a region R , oriented counterclockwise, and if the vector field $\vec{F}$ is differentiable on R , then

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \operatorname{curl}(\vec{F}) \, dA.$$

Equivalently,

$$\oint_C M \, dx + N \, dy = \iint_R (N_x - M_y) \, dA.$$

Example Using Green's Theorem

Warning: Green's theorem applies only to closed curves.

Let C be the circle of radius 1 centered at $(2, 1)$, oriented counterclockwise. Evaluate

$$\oint_C ye^{-x} \, dx + \left(\frac{1}{2}x^2 - e^{-x} \right) \, dy.$$

Direct Parametrization

A parametrization of the circle is

$$x = 2 + \cos \theta, \quad y = 1 + \sin \theta, \quad 0 \leq \theta \leq 2\pi.$$

Then

$$dx = -\sin \theta \, d\theta, \quad dy = \cos \theta \, d\theta.$$

Using Green's Theorem

Let

$$M(x, y) = ye^{-x}, \quad N(x, y) = \frac{1}{2}x^2 - e^{-x}.$$

Then

$$N_x = x + e^{-x}, \quad M_y = e^{-x}.$$

Hence

$$N_x - M_y = (x + e^{-x}) - e^{-x} = x.$$

By Green's theorem,

$$\oint_C M \, dx + N \, dy = \iint_R (N_x - M_y) \, dA = \iint_R x \, dA.$$

Using symmetry,

$$\iint_R x \, dA = \text{Area}(R) \cdot \bar{x},$$

where $\bar{x}$ is the x -coordinate of the center of the disk.

Since the circle has radius 1,

$$\text{Area}(R) = \pi,$$

and since the center is $(2, 1)$,

$$\bar{x} = 2.$$

Therefore,

$$\iint_R x \, dA = \pi(2) = 2\pi.$$

Thus,

$$\boxed{\oint_C ye^{-x} \, dx + \left(\frac{1}{2}x^2 - e^{-x}\right) \, dy = 2\pi}$$

Centroid and Conservative Fields

Recall that the x -coordinate of the centroid is

$$\bar{x} = \frac{1}{\text{Area}} \iint_R x \, dA.$$

More generally,

$$\bar{x} = \frac{1}{\text{mass}} \iint_R x \, \delta \, dA,$$

where δ is the density function.

Special Case

If

$$\text{curl}(\vec{F}) = 0,$$

then $\vec{F}$ is conservative.

By Green's theorem,

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl}(\vec{F}) \, dA.$$

Hence, if

$$\text{curl}(\vec{F}) = 0 \quad \text{in } R,$$

then

$$\oint_C \vec{F} \cdot d\vec{r} = 0.$$

Consequence

If $\vec{F}$ is defined everywhere in the plane and

$$\text{curl}(\vec{F}) = 0,$$

then $\vec{F}$ is conservative.

Proof Idea:

Apply Green's theorem to the vector field. Since the curl is zero, every closed line integral is zero, hence the field is conservative.

Proof of Green's Theorem

Proof Ideas for Green's Theorem

Recall Green's theorem:

$$\oint_C M dx + N dy = \iint_R (N_x - M_y) dA.$$

Observe that

$$\oint_C M dx = \iint_R -M_y dA.$$

Similarly,

$$\oint_C N dy = \iint_R N_x dA.$$

Combining these gives Green's theorem.

Decomposition into Simple Regions

We can decompose the region R into simple regions.

Suppose

$$R = R_1 \cup R_2.$$

Then

$$\oint_C M dx = \iint_{R_1} -M_y dA + \iint_{R_2} -M_y dA.$$

Hence,

$$\oint_C M dx = \oint_{C_1} M dx + \oint_{C_2} M dx.$$

The integrals along the common boundary cancel because the boundary is traversed twice with opposite orientations.

Thus, we may cut R into vertically simple regions

$$a \leq x \leq b.$$

Proof:

$$\oint_C M dx = \iint_R -M_y dA$$

If R is vertically simple and C is the boundary of R oriented counterclockwise.

Main Step

Prove

$$\oint_C M dx = \iint_R -M_y dA$$

If R is vertically simple.

$$\oint_C M dx = \int_a^b M(x, f_1(x)) dx$$

where

$$y = f_1(x), \quad x \text{ from } a \text{ to } b$$

$$\int_{C_2} M(x, y) dx$$

On C_2 ,

$$x = b, \quad dx = 0$$

Similarly,

$$\int_{C_4} M dx = 0$$

$$\int_{C_3} M dx = \int_b^a M(x, f_2(x)) dx = - \int_a^b M(x, f_2(x)) dx$$

where

$$y = f_2(x)$$

Therefore,

$$\oint_C M dx = \int_a^b M(x, f_1(x)) dx - \int_a^b M(x, f_2(x)) dx$$

Also,

$$\iint_R -M_y dA = - \int_a^b \int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dy dx$$

Inner integral:

$$\int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dy = M(x, f_2(x)) - M(x, f_1(x))$$

Hence,

$$\iint_R -M_y dA = \int_a^b [M(x, f_1(x)) - M(x, f_2(x))] dx$$

Finally,

$$\oint_C x dy = \iint_R 1 dA = \text{Area}(R)$$

Lecture 23: Flux Normal Form of Green's Theorem

Flux: another line integral.

- C : plane curve
- $\vec{F}$: vector field

Flux of $\vec{F}$ across C is

$$\int_C \vec{F} \cdot \hat{n} \, ds$$

where

$$\hat{n} = \text{unit normal vector}$$

If we break C into small pieces Δs ,

$$\text{Flux} = \lim_{\Delta s \rightarrow 0} \sum \vec{F} \cdot \hat{n} \, \Delta s$$

Work integral:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \hat{T} \, ds$$

where $\hat{T}$ is the unit tangent vector.

- Work uses the tangential component of $\vec{F}$.
- Flux uses the normal component of $\vec{F}$.

$$\text{Flux} = \int_C \vec{F} \cdot \hat{n} \, ds$$

Interpretation of Flux

If $\vec{F}$ is a velocity field, then flux measures how much fluid passes through the curve C per unit time.

What passes through a portion Δs in unit time:

$$\text{Area} = \text{base} \times \text{height}$$

$$= \Delta s (\vec{F} \cdot \hat{n})$$

where $\hat{n}$ is the unit normal vector.

What flows across C :

- left to right is counted positively,
- right to left is counted negatively.

Example

Let C be the circle of radius a centered at the origin, oriented counterclockwise.

$$\vec{F} = x\hat{i} + y\hat{j}$$

Then

$$\vec{F} \cdot \hat{n} = |\vec{F}| \cdot \frac{1}{a} = a$$

$$\oint_C \vec{F} \cdot \hat{n} \, ds = \oint_C ds = \text{length}(C) = 2\pi a$$

Same C , $\vec{F} = \langle -y, x \rangle$

$\vec{F}$ tangent to C

$$\vec{F} \cdot \hat{n} = 0$$

Calculation using components

Remember

$$d\vec{r} = \vec{T} \, ds = \langle dx, dy \rangle$$

$\hat{n} = \hat{T}$ rotated 90° clockwise

$$\hat{n} \, ds = \langle dy, -dx \rangle$$

$$\hat{n} \, \Delta s = \langle \Delta y, -\Delta x \rangle$$

So, if

$$\vec{F} = \langle P, Q \rangle$$

then

$$\begin{aligned} \oint_C \vec{F} \cdot \hat{n} \, ds &= \oint_C \langle P, Q \rangle \cdot \langle dy, -dx \rangle \\ &= \oint_C (-Q \, dx + P \, dy) \end{aligned}$$

If

$$\vec{F} = \langle M, N \rangle$$

then

$$\oint_C \vec{F} \cdot \hat{n} \, ds = \oint_C (-N \, dx + M \, dy)$$

Green's theorem for flux

If C encloses a region R in counterclockwise orientation, then

$$\oint_C \vec{F} \cdot \hat{n} \, ds = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dA$$